Calorie Predictor Final Report

DS3000

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Abstract

Our project aimed to predict the amount of calories within foods based on the macronutrients that the specific food contained. Using ML models and data using macronutrients within foods, we were able to predict the amount of calories that would result from those specific meals. Using the data we learned that there are strong linear relations between calorie counts and macronutrients (more specifically fats and carbohydrates). Using the linear regression model, we found that the model met the core assumptions reasonably well as residuals displayed no strong patterns, variance appeared nearly constant, and QQ-plots showed near-normal errors with a mild skew. The analysis displayed that calories can be reliably predicted from the macronutrient composition of foods using regression techniques.

Introduction

As social media continues to grow, so do portrayed unrealistic expectations of physical standards and living standards. Food and cooking are both large parts of many social media platforms which can influence many viewers into changing their eating habits or lifestyle habits into a more unhealthy direction. Taking this worry into account, and as our motivation for our project, we wanted to focus on proving how healthy food items are calorically given the nutrient information. By analyzing this, we are able to understand what is more influential on an individual's caloric intake as we can also find the cause of it by tracing back to the food item's nutrients. Listed below are some of the key questions:

1. How influential are specific food items within certain categories (proteins, fibers, lipids) towards a user's diet?

2. What noticeable patterns within the data cause different food options to create such stark nutritional benefits?

Data Description

Our data primarily comes from the USDA FoodData Central API (https://api.nal.usda.gov/fdc/v1/foods/search). This provides lots of information about the macronutrients of any foods available. The API returns data on all the different stats such as the vitamins, minerals, macros for each food item.

Summary of the data processing pipeline:

1. Query the USDA food API
2. Clean the data to prepare the data frame for visualization and analysis
3. Visualize using Seaborn and Matplotlib
4. Implement Machine Learning models and perform regression analysis

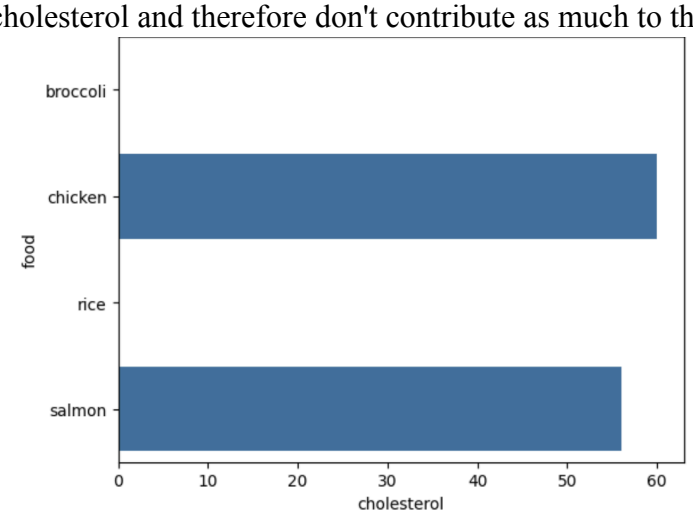
To begin analyzing the data, implemented the function get_macros(food) which queries the USDA API for food items. We collected our data on a sample recipe of chicken, salmon, broccoli and rice. For our third model, we expanded our dataset to have beef, pork, lamb. Each API call gets 50 foods and their nutritional values.

The raw API data needed a lot of cleaning in order for it to be analyzed. Using the clean_macros() function, we did the following: column standardization, feature selection, data type conversion and removed NaN values. In order to deal with outliers, we used the IQR using the method remove_outliers_iqr() to remove outliers that could potentially mess with the final data result. After cleaning, our dataset contained the nutrition data for an item's protein, fat, carbs, sugars, sodium and cholesterol.

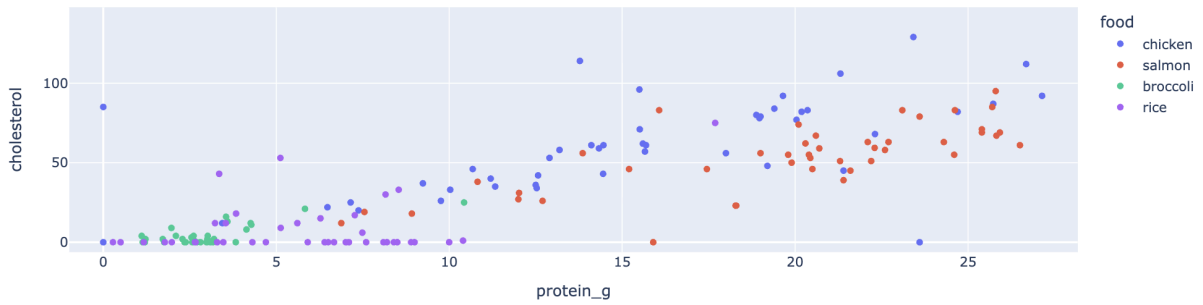
food	description	category	category_list	protein_g	fat_g	carbs_g	calories	sugar_g	fiber_g	iron	sodium	cholesterol	saturated_fats	added_sugars
chicken	CHICKEN	Canned Meat	[Canned Meat]	21.40	1.79	0.00	107.0	0.00	0.0	0.00	179.0	45.0	0.89	0.0
chicken	CHICKEN	Canned Soup	[Canned Soup]	6.67	0.00	33.30	167.0	13.30	0.0	0.00	5400.0	0.0	0.00	0.0
chicken	CHICKEN	Poultry, Chicken & Turkey	[Poultry, Chicken & Turkey]	24.70	9.41	1.18	188.0	1.18	0.0	0.42	353.0	82.0	2.94	0.0
chicken	CHICKEN	Pepperoni, Salami & Cold Cuts	[Pepperoni, Salami & Cold Cuts]	15.60	15.60	9.38	219.0	3.12	0.0	1.12	1160.0	62.0	4.69	0.0
chicken	CHICKEN	Seasoning Mixes, Salts, Marinades	[Seasoning Mixes, Salts, Marinades &	0.00	0.00	111.00	556.0	0.00	0.0	0.00	0.0	0.0	0.00	0.0

The example of cleaned data for chicken

We then visualized our data with several plots. The first plot was a bar plot that grouped the median cholesterol by food type. It showed that chicken and salmon had a lot more cholesterol and contribute more to the overall result in a recipe while rice and broccoli don't have as much cholesterol and therefore don't contribute as much to the overall cholesterol.



We also made a scatter plot comparing the protein to cholesterol, showing a positive relationship between protein and cholesterol.



Scatterplot comparing protein to cholesterol of each food type

Methods

To determine if the calories can be accurately predicted given a food's macronutrient data, we designed a modeling pipeline based on mathematical assumptions of supervised learning and more specifically regression-based models. The research question that we looked into looked at: "Can caloric content be predicted using macronutrients such as protein, carbohydrates, and fat?" Based on this, we curated a nutritional dataset and created models using linear regression and polynomial regression. We focused specifically on cleaning the dataset in order to prioritize the nutrients listed within the question. What we wanted to discover was the significance of these three macronutrients and the caloric result.

In regards to the data cleaning and preparation, before conducting any modeling, we made sure to clean the data to ensure the integrity of our predictors. To do this, we standardized nutrient columns and removed non-numerical/irrelevant fields, identified and dealt with missing values, removed duplicate items and consolidated certain categories, and ensured consistency in nutrient units throughout all entries. The resulting cleaned dataset stored the core macronutrient information (proteins, carbohydrates, fats), alongside more supplementary nutrients (fiber, sugars, cholesterol) which can be somewhat influential towards calorie count.

Another way that we improved our model was by applying the Interquartile Range (IQR) method to find and remove outliers from the key variables. Given the wide variety of food items within the dataset, it comes with extreme values. By getting rid of these outliers, we were able to focus the model on a more consistent relationship between macronutrients and calories. In our linear regression model, we returned the mean squared error (MSE) and the R^2 score. In the model we took in the protein (protein_g), fat (fat_g), and carbohydrates (carbs_g) as we found them most influential on calorie amount. For our second model, we decided to use a polynomial regression instead of a linear regression to see if it would help with our results. We then also tried a third model using a linear regression on a subclass of red meats instead of a general recipe, to see if we could predict the protein content based on the cholesterol.

Results

Model 1 Linear Regression:

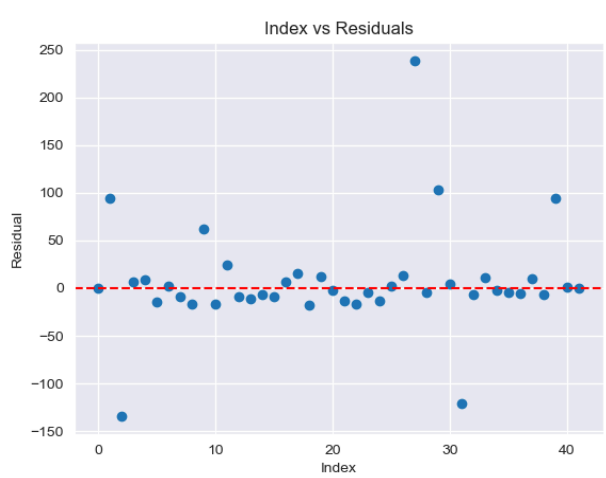
Our baseline linear regression model predicted the calories from protein, fat and carbohydrates. It gave a mean squared error of 2996.54 and an R^2 of 0.437. The R^2 value of 0.437 indicates that our model explains about 44% of the variance in the calorie content based only on the macros of the foods. This represents predictive power and it does much better than random guessing. The MSE of 2996 shows an average error of about 54.7 calories for each food. In order to assess whether our linear regression model is valid, we examined Independence, Constant Variance/Linearity and Normality.

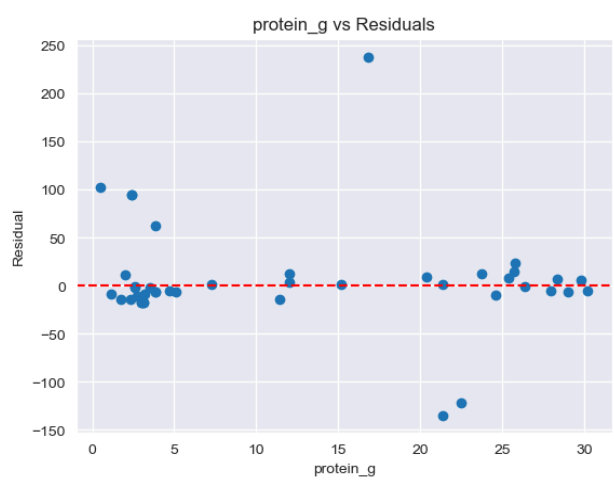
The "Index vs. Residuals" showed scattered points mainly around the zero without any clear patterns or trends. In addition to that, there are no clear clusters or correlations which are apparent in the graph. Therefore this most likely points that this satisfies the independence test.

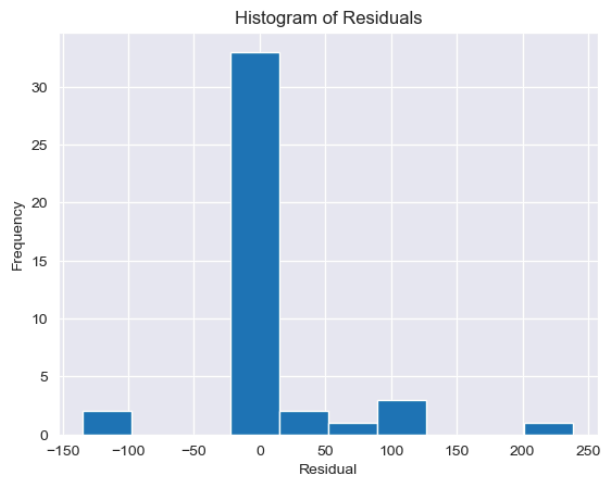
The "protein_g vs. Residuals" plot shows mainly that the residuals are randomly scattered around zero, and that the spread appears to be consistent across the entire range of protein_g. There isn't any "fanning" patterns which would have indicated heteroscedasticity. The random scatter throughout the zero supports the linearity assumption for the data not including some of the major outliers.

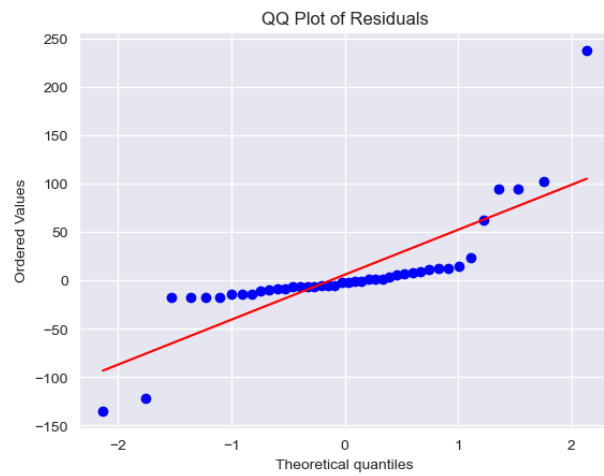
The histogram points that there is a heavy concentration around the 0 and that there are also clear outliers on both tails. The QQ plot of residuals shows that there is a good fit in the middle range which includes the points that follow the line but there are more outliers on the upper tail of the graph. Normality doesn't seem to hold as the points aren't clearly following the diagonal line seen in the QQ plot.

Overall, the plots show that the independence and constant variance are reasonable, but normality is violated due to many outliers. Combined with a below average R^2, this suggests that while the model does capture some relationship between protein and calories, it is not complete, and needs some additional predictors.









Model 2 Polynomial Regression:

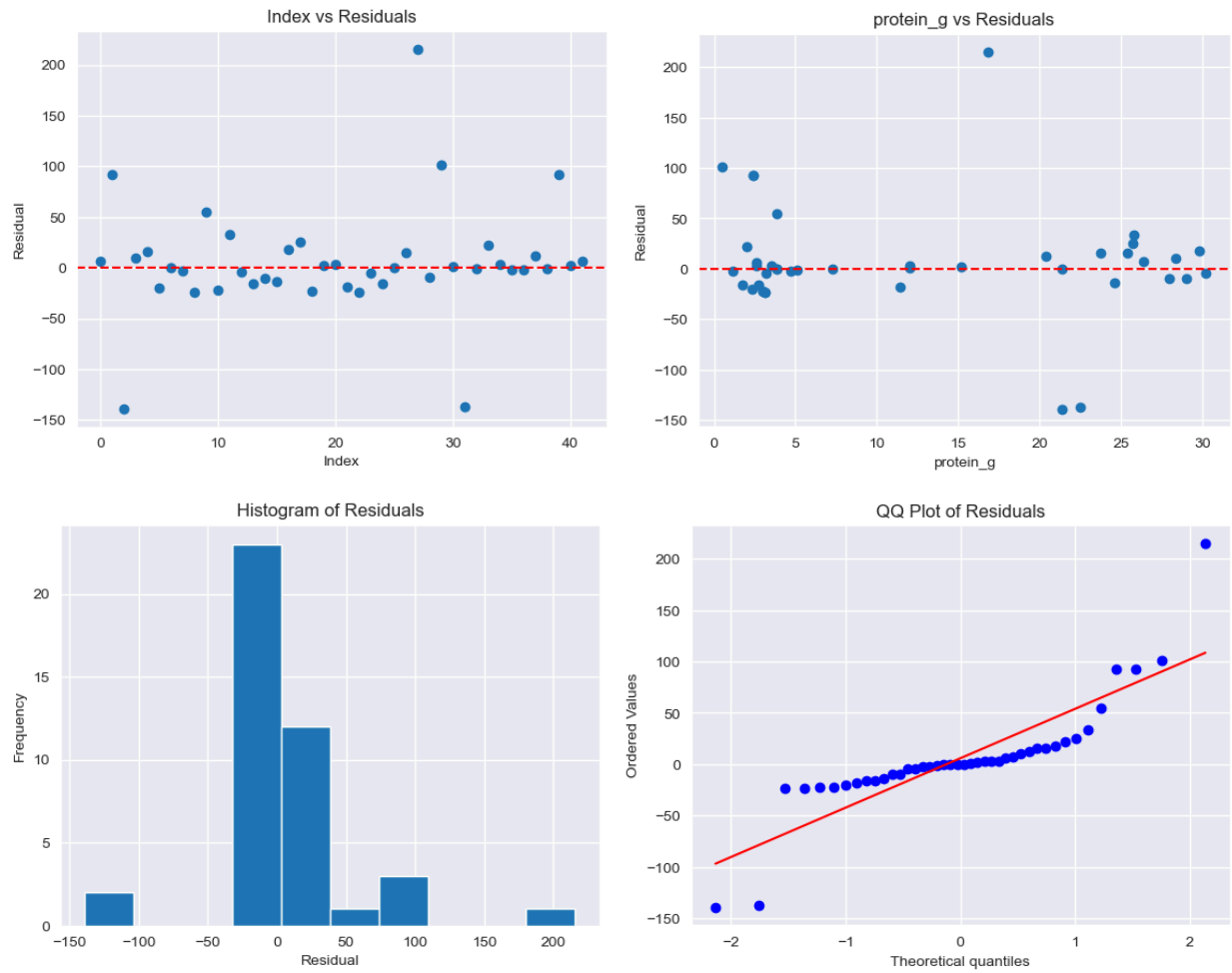
For our second model we used a polynomial regression in an attempt to better the results of model 1. Specifically, we used the formula $fat + carbs + protein + fat^2 + fat * carbs + protein^2$ under the suspicion that grams of fat and protein contribute greater to total calories than carbs. However, the results we got was an R^2 of 0.454 and a MSE of 2906. This means our model accounts for roughly 45% of the variance of calorie count based on the macros and that our average error was 53.9 calories per food item.

In order to assess whether our linear regression model is valid, we examined Independence, Constant Variance/Linearity and Normality of the residuals.

What was found was almost the same results as with model 1:

Index vs residuals mostly around 0 with few outliers suggesting independence. Protein vs residuals mostly around 0 with no clear patterns indicating homoscedasticity. A histogram of residuals that shows normal shape to the distribution and a QQ plot that shows that there are more residuals around 0 then world be in a normal distribution.

Overall, the residuals indicate good independence but leave a bit to be decided in terms of normality.



Model 3:

For the third model we decided to pivot our focus to see if we could predict the grams of protein in a given food based on the cholesterol. To do this we employed a simple linear regression model and got as a result a R^2 of 0.521 and a MSE of 23.7. This means our model explained roughly 50% of the variance and was only off by an average of 4.9 grams of protein per food item.

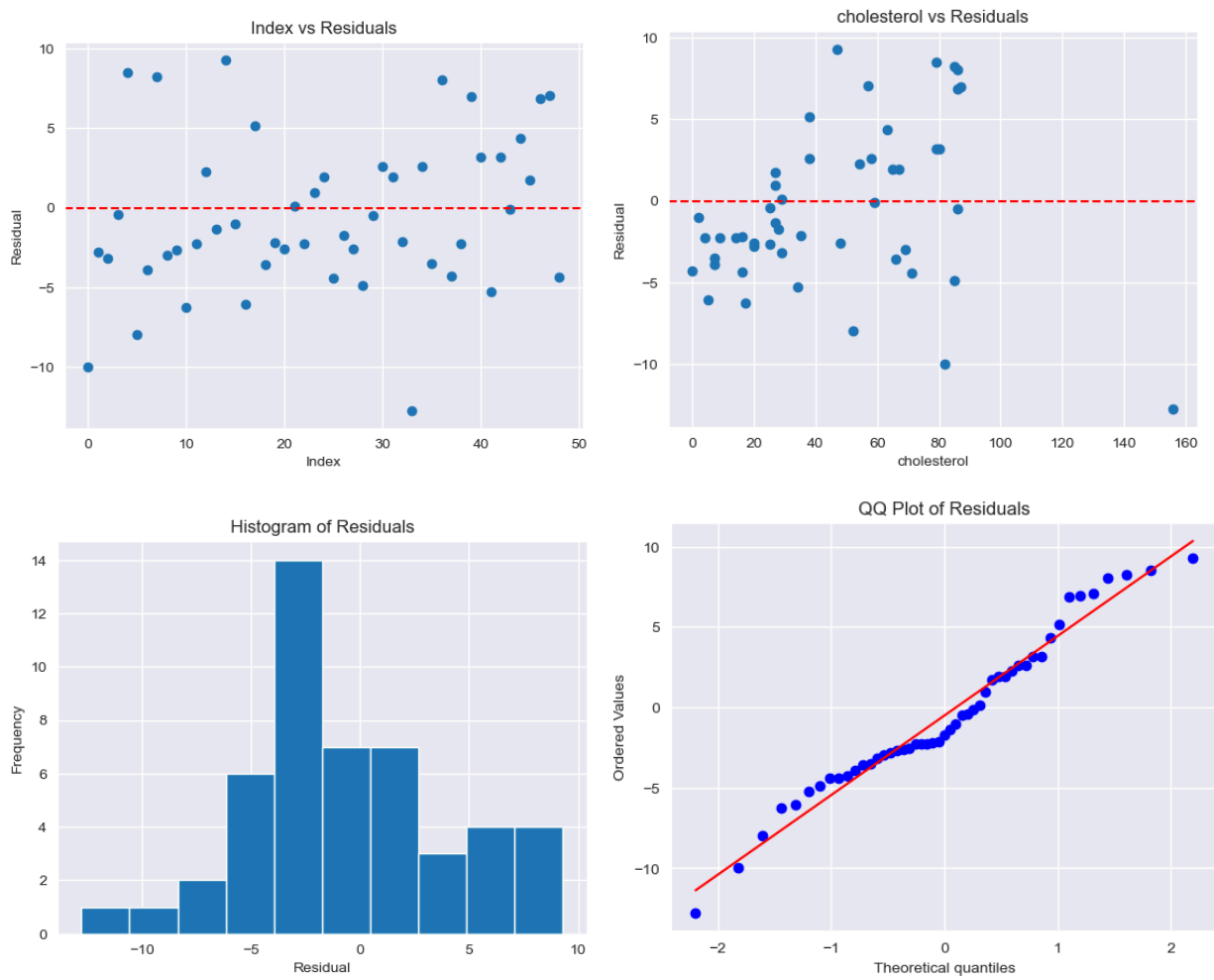
Residuals:

From the index vs residuals plot there is a seemingly random scattering around 0 with some noticeable outliers. There doesn't seem to be much of any pattern suggesting independence.

However, in cholesterol vs residuals there seems to be a fanning effect that suggests heteroscedasticity where residuals spread out the higher protein gets. What this means is that our model may lose its accuracy the higher cholesterol gets.

As for the distribution, the histogram and the QQ plot showed a normal distribution with only a slight right skew which was a promising sign for the residuals.

Overall, the initial results from this model were promising but the residuals seem to highlight problems with the potential viability when cholesterol increases.



Discussion

The findings provided show both the success and remaining complexities for predicting caloric content from the macronutrient composition Though our linear regression model showed strong results with an R^2 value of 0.437, the analysis shows both the strengths and future areas of refinement.

Though the linear regression model performed well, the diagnostic plots showed some insights about the assumption compliance. The independence assumption appeared satisfied evidenced by the random scatter without clustering or autocorrelation. Similarly, the constant variance and linearity assumptions were both reasonably met with residuals being scattered randomly around

zero across the range of predictor values. On the other hand, the normality assumption showed violations especially in the tails of the distribution. The QQ plot showed that though the central portion of residuals followed normal distribution, both the upper and lower tails deviated from normality showing that there still remained a presence of outliers. Looking into these outliers, they are more likely food items with unnatural nutrient compositions (extremely processed foods/heavily concentrated foods). Though normality was violated, it doesn't undermine the model's validity.

Comparative Model Analysis

In the second model, we found more of the same, finding incredibly similar results in terms of performance and residuals. While this may seem surprising to see a different model perform so similarly, it makes sense if you consider that the linear regression may have found negligible coefficients for the higher order terms, leaving only the linear terms. What this suggests is that our attempt to better the model by adding higher order polynomial terms just resulted with the regression emphasizing the linear terms and recreated the model 1 linear regression. In the end this is an indicator to us that the model is better modeled linearly than polynomially. If anything the comparison between models suggests that for further improvements it may be beneficial to stick with the linear model and add predictors in an attempt to account for the remaining variance especially with such stark outliers suggesting there is something there we aren't considering.

Looking at the estimated 56% variance which remained unexplained by the model can be taken into consideration. Currently our model doesn't look at alcohol content which has a large calorie content per gram. In addition, food processing methods like cooking or fermentation can affect the digestible energy content of foods which can overall affect the macronutrient measurements. Lastly, there could be remaining issues in the dataset where values for macronutrients are estimated rather than directly measured.

Despite these listed limitations, our model does demonstrate practical use. With a prediction error of 71 calories, we find it to be sufficiently accurate for dietary planning and nutritional analysis. In addition, this model is practical because the amount of variables which influence the result (protein, fats, carbs) are all available on nutritional labels.

Future Research Directions:

- Conducting analysis to examine which food categories produce the largest prediction errors could identify specific domains where the linear model breaks down. An example of this would include fried, baked, fermented, or pickled goods which could require category-specific adjustments.
- We could also expand the model to include fiber content, alcohol content, and more food processing indicators which could help us capture more variance and improve model predictions for food subcategories.
- We could also include feature engineering for processed foods by creating categorical indicators for processing levels (minimal processing, moderate processing, high processing).

As for the protein prediction model we saw good potential with the model but notably face the issue of heteroscedasticity. This might be explained by the fact that we only have one predictor but might also just be the product of the linear model. As for future direction the model could potentially be changed to something other than linear or more predictors could be added and then a PCA test run to see which ones truly matter.

In the end, this third model still seems to be useful in predicting protein content in foods with lower cholesterol which could help people manage their protein intake and make informed decisions on their protein intake even if they only have cholesterol data. Furthermore it can help inform people about the tradeoff between the two and how more cholesterol (typically seen as bad) correlates to more protein (typically seen as good).

In conclusion, we created a solid model as a basis and can expand upon it in the future. Future implementations of this model can focus more specifically on separate categories or focus on adding additional variables which can have an effect on the total caloric amount. The current model is also relevant as the R^2 value shows a strong relationship between calories and the selected macronutrients. Through this research we also learned extensively of the impact of separate macronutrients on one's diet and lifestyle.